

Debugging.

Failure Class.

↳ It is the category of defect responsible for an incorrect agent response.

Common langchain agent failure classes.

1. Wrong tool selection.

2. Missing tool call.

3. Bad tool arguments.

4. Weak Retrieval.

5. Hallucinated final response.

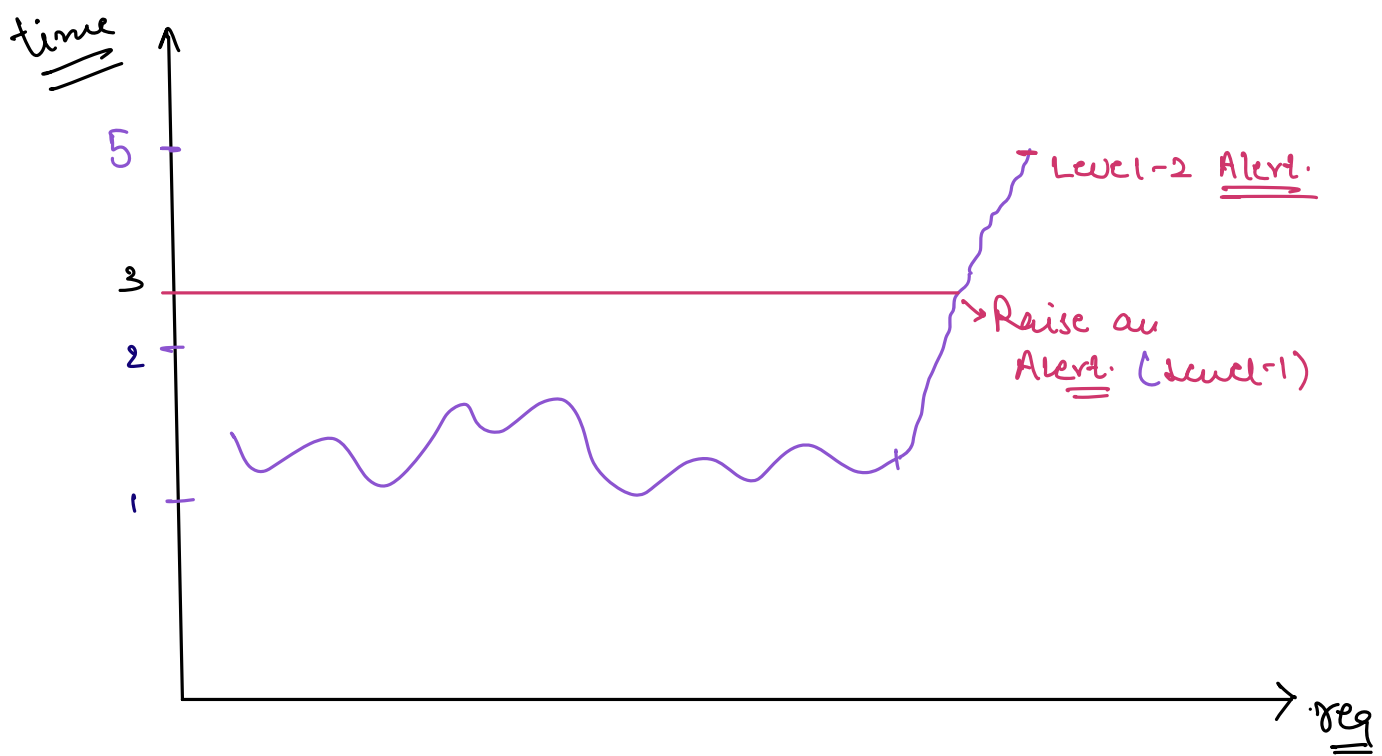
6. Over Refusal | Under Refusal

↳ Agent refusing even the queries which were supposed to answer.

↳ Agent answers unsupported / out-of-domain queries.

7. looping / Excessive tool call.

8. Slow response.



Observe

- Time
- Memory
- No. of failures.

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Debugging : final answer (vs) trajectory

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The path our Agent
followed.

Failure class → remediation strategy

Failure class	Best first fix	📄
Wrong tool selection	Improve tool name, description, examples, and system prompt	
Missing tool call	Add explicit "must use tool when..." rule	
Bad tool arguments	Improve tool schema and input examples	
Weak retrieval	Tune <code>k</code> , chunk size, metadata filters, query rewriting	
Hallucination	Add grounding rule: answer only from retrieved/tool data	
Over-refusal	Relax refusal rule and improve "answer when enough evidence exists"	
Under-refusal	Add stricter unsupported-query detection	
Excessive tool calls	Add max iterations, better stopping rules, tool call limits	
High latency	Reduce retrieved context, reduce tool calls, cache stable results	
High cost	Reduce prompt size, context size, model calls, and unnecessary tools	

Prompt Patch.

Tool Patch.

Retrieval Time.

Quality Metrics

Cost (vs) Latency Trade-off.